

Tutku Altınyaprak

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EDUCATION

Haliç University

Bachelor of Science in Computer Engineering; GPA: 3.79/4.00

Istanbul, Turkey

Sep. 2022 — Expected Summer 2027

EXPERIENCE

Backend Developer

Google Developer Groups (GDG)

Sep. 2025 – Present

Istanbul, Turkey

- Collaborate within a dynamic team to brainstorm and design backend architectures for various community-driven software projects.
- Contribute to the feature ideation and system design of upcoming open-source initiatives and real-time interactive platforms.

Backend Developer Intern

Tezmaksan Makina San. ve Tic. A.Ş.

Jun. 2025 – Aug. 2025

Istanbul, Turkey

- Engineered a high-performance CNC monitoring service in C++ from scratch as an R&D initiative, replacing a legacy C# system and achieving significantly better resource optimization.
- Designed a multi-threaded architecture to concurrently read and process raw and calculated data from multiple CNC machines, ensuring real-time data flow.
- Integrated robust SQL database logging for processed machine metrics and successfully validated the entire system in a live factory environment.
- Conducted software testing and authored comprehensive technical documentation to facilitate the company-wide adoption of the newly developed C++ service.

PROJECTS

NeuroPlan (Work Management System) | C#, ASP.NET Core, Python, Scikit-learn

Feb. 2026 – Present

- Architecting a scalable work tracking system utilizing Pragmatic Clean Architecture.
- Integrating an ML and NLP-to-SQL assistant for intelligent task management insights.
- Developing RESTful APIs with ASP.NET Core, preparing for React frontend integration.

Real-Time CNC Monitoring Service | C++, Multi-threading, WebSockets, SSE, SQL

Jun. 2025 – Aug. 2025

- Engineered a high-performance backend from scratch to process real-time CNC telemetry.
- Implemented robust thread management and SSE protocols for concurrent data streaming.
- Validated the system in a live factory environment, replacing a legacy infrastructure.

5G Network Resource Optimizer | Python, FastAPI, Scikit-learn, JavaScript

Late 2025

- Developed a Scikit-learn ML model to optimize 5G network resource allocation.
- Designed a Human-in-the-Loop system with manual logging for continuous refinement.
- Co-authored a manuscript on the architecture, published in Springer's *Discover Sensors*.

Enterprise Leave Tracking System | C#, WPF, SQL Server

Early 2025

- Developed a desktop application (WPF, C#) for efficient employee leave management.
- Commercialized and sold the software to an industrial firm for active daily use.

TECHNICAL SKILLS

Languages: C#, C++, Python, SQL, JavaScript, HTML/CSS

Frameworks: ASP.NET Core, FastAPI

Developer Tools: Git, GitHub, Postman, Visual Studio, VS Code, Docker

Architecture & Concepts: RESTful APIs, Clean Architecture, N-Tier Architecture, Machine Learning